
Margin Requirements

Formula Reference Document

Trade Republic — Margin Trading Methodology

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This document sets out all quantitative formulas used in Trade Republic's Margin Requirements Methodology. Formulas are presented in the order in which they appear in the methodology document, grouped by component. Variable definitions accompany each formula block. A consolidated formula reference appears in the Appendix.

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1. Margin (Section D)

1.1 D.1 — Four Layers of Margin

The margin requirement framework consists of four layers applied on an instrument-by-instrument basis:

1. **Initial Margin (IM)** — minimum equity to open a leveraged position.
2. **Maintenance Margin (MM)** — minimum equity the customer must maintain at all times after a position is opened.
3. **Variation Margin (VM)** — daily cash P&L settlement applied to all open positions subject to mark-to-market valuation.
4. **Overnight Maintenance Margin** — additional haircut-based collateral requirement applied at end-of-day to account for the inability to monitor positions outside of trading hours.

1.1.1 D.1.1 — Initial Margin Rate

The initial margin rate is the minimum equity the customer must post before a leveraged position can be opened. It is calibrated at instrument level using a scenario-based Value-at-Risk methodology and is expressed as a fraction of the notional position value. The rate is composed of two additive components: a Market Risk Component (MRC), which captures the potential loss over the assumed liquidation horizon at a 99% confidence level, and a Liquidity Component (LC), which accounts for the cost of unwinding the position in adverse market conditions.

Initial Margin Rate

$$\text{IM}_{\text{rate}}(i) = \text{MRC}(i) + \text{LC}(i)$$

Market Risk Component

$$\text{MRC}(i) = \max\left(\text{RVaR}^{\text{hist}}(i) + \text{CB}(i), w_s \cdot \text{RVaR}^{\text{stress}}(i)\right)$$

Liquidity Component

$$\text{LC}(i) = \text{DF} \cdot \text{RVaR}^{\text{hist}}(i) \cdot \lambda(i)$$

Symbol	Description
$\text{IM}_{\text{rate}}(i)$	Initial margin rate for instrument i as a fraction of position value
$\text{MRC}(i)$	Market Risk Component for instrument i
$\text{LC}(i)$	Liquidity Component for instrument i
$\text{RVaR}^{\text{hist}}(i)$	Robust VaR from filtered historical scenarios (see Section D.2.4)
$\text{RVaR}^{\text{stress}}(i)$	Robust VaR from stressed scenarios (see Section D.2.2)
$\text{CB}(i)$	Correlation Break adjustment (see Section D.2.4)
w_s	Stress weight (default: 0.5)
DF	Diversification factor for the liquidation group
$\lambda(i)$	Days-to-liquidate scaling factor for instrument i

The initial margin rate is computed on a standalone basis per instrument, meaning each instrument is treated as its own liquidation group. This ensures that the rate displayed to the customer prior to trade execution reflects the true cost of holding that specific instrument on leverage, without benefit from diversification across instruments the customer does not hold. Full details of the calibration methodology are provided in Section D.2.

1.1.2 D.1.2 — Maintenance Margin Rate

The maintenance margin rate represents the minimum equity ratio the customer must maintain at all times after a leveraged position is opened. It is expressed as a minimum Adjusted Collateral Value relative to the outstanding loan, governed by the LTV monitoring threshold $\overline{\text{LTV}}$. For each instrument i , the required minimum ACV at time t is:

Minimum Required ACV

$$\text{ACV}_t^{\min} = \frac{L_t}{\overline{\text{LTV}}}$$

Maintenance Margin Rate

$$\text{MM}_{\text{rate}}(i) = 1 - \overline{\text{LTV}} \cdot (1 - h_{\text{eff}}(i))$$

Symbol	Description
$ACV_t^{\min}$	Minimum Adjusted Collateral Value required at time t
L_t	Outstanding loan balance at time t
$\overline{LTV}$	LTV monitoring threshold (default: 75%)
$h_{\text{eff}}(i)$	Effective haircut for instrument i (see Section E.2)
$MM_{\text{rate}}(i)$	Maintenance margin rate for instrument i as a fraction of market value

The maintenance margin rate is instrument-specific: instruments with higher effective haircuts require a larger minimum equity cushion for the same loan amount. A customer whose ACV falls below $ACV_t^{\min}$ — equivalently, whose LTV exceeds $\overline{LTV}$ — is subject to the escalation framework described in Section D.1.4.

Two-Phase LTV Framework. The margin framework operates in two distinct phases tied to the liquidation horizon T_{liq} (default: 4 trading days):

- **Phase 1 — Opening ($t = 0$):** At trade inception the customer must post Initial Margin, so the opening LTV is directly determined by the IM rate:

$$LTV_0 = \frac{L_0}{ACV_0} = \frac{1 - IM_{\text{rate}}(i)}{1 - h_{\text{eff}}(i)}$$

- **Phase 2 — Seasoned ($t > T_{\text{liq}}$):** Once the position has been held beyond the liquidation horizon, the Initial Margin requirement no longer applies. The binding constraint becomes the maintenance margin threshold $\overline{LTV} = 75\%$, monitored continuously via daily Variation Margin and end-of-day Overnight Maintenance Margin. The outstanding loan evolves as:

$$L_t = L_0 - \sum_{\tau=1}^t VM_{\tau}$$

and a margin event is triggered whenever $LTV_t > \overline{LTV}$.

This two-phase structure ensures that the Initial Margin covers the worst-case loss over the full liquidation horizon, while the lighter Maintenance Margin framework governs the ongoing life of the position once that horizon has elapsed.

1.1.3 D.1.3 — Variation Margin

The profit or loss arising from the change in the theoretical value of an open position between two consecutive trading days. Gains and losses are settled symmetrically and variation margin is not held as collateral.

Variation Margin

$$VM_{i,t} = (P_{i,t} - P_{i,t-1}) \times N_i$$

Positive VM is credited to the customer; negative VM is debited. Applied to all positions on every trading day.

Symbol	Description
$VM_{i,t}$	Variation Margin for instrument i on day t
$P_{i,t}$	Settlement price of instrument i on day t
$P_{i,t-1}$	Settlement price of instrument i on the previous trading day
N_i	Quantity held of instrument i

1.1.4 D.1.4 — Overnight Maintenance Margin

At the close of each trading day, Trade Republic applies an overnight maintenance margin requirement to all open leveraged positions, reflecting the increased risk arising from the inability to monitor or act on adverse price movements outside of trading hours. The overnight margin is expressed as a minimum Adjusted Collateral Value relative to the outstanding loan, calibrated using a one-day parametric Value-at-Risk at the 99% confidence level, scaled to the overnight holding period. Specifically, the overnight haircut applied to each instrument is derived as:

Overnight Haircut

$$h_{\text{overnight}}(i) = h_{\text{floor}}(\text{class}(i)) + w_{\text{buf}} \cdot \max(0, h_{\text{vol}}(i) - h_{\text{floor}}(\text{class}(i)))$$

where h_{floor} is the asset-class minimum haircut, h_{vol} is the volatility-implied haircut based on the instrument's EWMA historical volatility, and w_{buf} is a volatility buffer weight. This ensures that the

collateral discount applied overnight is anchored to a regulatory-grade asset class floor while remaining responsive to current market conditions. Any customer whose LTV breaches the monitoring threshold of 75% at end-of-day is subject to an overnight margin review, with corrective action — either additional collateral posting or partial position reduction — required prior to the opening of the next trading session.

Symbol	Description
$h_{\text{overnight}}(i)$	Overnight haircut for instrument i
$h_{\text{floor}}(\text{class}(i))$	Asset-class minimum haircut floor
$h_{\text{vol}}(i)$	Volatility-implied haircut based on EWMA historical volatility
w_{buf}	Volatility buffer weight ($0 \leq w_{\text{buf}} \leq 1$)

1.1.5 D.1.5 — Monitoring Framework

The monitoring framework operates in two distinct phases determined by the liquidation horizon T_{liq} (default: 4 trading days). The set of active checks and the associated breach actions differ between the two phases.

Phase 1 — Within Liquidation Horizon ($0 < t \leq T_{\text{liq}}$). During this phase the Initial Margin buffer is still the primary safeguard. Daily Variation Margin is applied and the LTV is monitored continuously. A breach signals that adverse price moves have consumed the IM buffer faster than the position can be wound down.

Check	Condition	Action
IM + VM intraday	$LTV_t = L_t / ACV_t > \overline{LTV}$	Monitoring alert
IM + VM end-of-day	$LTV_t^{\text{overnight}} > \overline{LTV}$	Overnight margin review
Liquidation	$LTV_t > LTV_{\text{liq}}$	Auto-liquidate

where the loan evolves daily as:

$$L_t = L_0 - \sum_{\tau=1}^t VM_{\tau}$$

and $LTV_t^{\text{overnight}}$ uses the overnight haircut $h_{\text{overnight}}(i)$ (Section D.1.4) in place of $h_{\text{eff}}(i)$ when computing ACV_t .

Phase 2 — Seasoned Position ($t > T_{\text{liq}}$). Once the position has been held beyond the liquidation horizon, the Initial Margin requirement is no longer binding. The lighter Maintenance Margin framework applies, supplemented by the overnight check at each close of business.

Check	Condition	Action
VM intraday	$LTV_t > \overline{LTV}$	Monitoring alert
MM end-of-day	$LTV_t^{\text{overnight}} > \overline{LTV}$	Overnight margin review
Liquidation	$LTV_t > LTV_{\text{liq}}$	Auto-liquidate

Escalation Logic. The three breach conditions are ordered by severity and share the same LTV thresholds across both phases:

Level	Trigger	LTV Threshold	Description
1 — Monitoring	$ACV_t < L_t / \overline{LTV}$	75%	Enhanced surveillance; no immediate customer action required
2 — Overnight breach	$ACV_t^{\text{overnight}} < L_t / \overline{LTV}$	75% (overnight ACV)	Customer must post collateral or reduce loan before next open
3 — Liquidation	$ACV_t < L_t / LTV_{\text{liq}}$	90%	Positions auto-liquidated to repay outstanding loan

Symbol	Description
T_{liq}	Liquidation horizon (default: 4 trading days)
L_t	Outstanding loan at time t , net of accumulated VM
ACV_t	Adjusted Collateral Value at time t using intraday haircut h_{eff}
$\text{ACV}_t^{\text{overnight}}$	Adjusted Collateral Value at close using overnight haircut $h_{\text{overnight}}$
$\overline{\text{LTV}}$	Monitoring threshold (default: 75%)
LTV_{liq}	Liquidation threshold (default: 90%)

1.2 D.2 — Calibration of Margin Rates

Margin rates are calibrated at instrument level using a Value-at-Risk (VaR) methodology designed to estimate the loss that could arise during the assumed liquidation horizon at the chosen confidence level.

1.2.1 D.2.1 — Mapping Liquidation Groups

Every instrument is assigned to a liquidation group l whose members share sufficient characteristics — market, asset class, and risk profile — to be auctioned together. Each liquidation group may be further divided into liquidation group *splits*.

1.2.2 D.2.2 — Scenario Construction

Two distinct scenario sets are constructed to generate hypothetical future prices for every instrument.

Filtered Historical Scenarios. The first set consists of 750 observations drawn from recent market history. For each historical date s , the realised return over the relevant holding period is rescaled to reflect current volatility conditions — ensuring that calm historical periods do not produce an underestimate of today's risk:

Filtered (Volatility - Scaled) Return

$$r_{\text{filt}}(i, s) = r_{\text{hist}}(i, s) \times \frac{\hat{\sigma}_{i,0}}{\hat{\sigma}_{i,s}}$$

Filtered Historical Scenario Price

$$P_{\text{filt}}(i, s) = P_{i,0} \times (1 + r_{\text{filt}}(i, s))$$

Symbol	Description
$r_{\text{hist}}(i, s)$	Historical return of instrument i over the holding period ending on date s
$\hat{\sigma}_{i,0}$	Current volatility estimate for instrument i (at $T = 0$)
$\hat{\sigma}_{i,s}$	Historical volatility estimate for instrument i at the time of observation s
$P_{i,0}$	Current market price of instrument i (at $T = 0$)
$P_{\text{filt}}(i, s)$	Filtered scenario price for instrument i under scenario s

Stressed Scenarios. The second set consists of 250 scenarios drawn from identified stress periods in market history (exceptional volatility, sharp correlation dislocations, extreme price movements). Unlike the filtered scenarios, stressed returns are applied *without* any volatility adjustment:

Stressed Scenario Price

$$P_{\text{stress}}(i, s) = P_{i,0} \times (1 + r_{\text{stress}}(i, s))$$

Symbol	Description
$r_{\text{stress}}(i, s)$	Raw historical return of instrument i from a stress period, scenario s
$P_{\text{stress}}(i, s)$	Stressed scenario price for instrument i under scenario s

1.2.3 D.2.3 — Profit and Loss Distributions

For each scenario, position-level P&L is aggregated across instruments within a liquidation group split to produce the P&L distribution used for VaR estimation.

Instrument P&L under Scenario s

$$\text{PL}(i, s) = (P_{i,0} - P_{\text{scenario}}(i, s)) \times N_i$$

P_{scenario} is either P_{filt} or P_{stress} depending on the scenario sub-sample.

Liquidation Group Split P&L

$$\text{PL}(l, s) = \sum_{i \in l} \text{PL}(i, s)$$

Symbol	Description
$\text{PL}(i, s)$	P&L of instrument i under scenario s
$\text{PL}(l, s)$	Portfolio P&L for liquidation group split l under scenario s
N_i	Quantity held of instrument i
$i \in l$	All instruments assigned to liquidation group split l

1.2.4 D.2.4 — Market Risk Component

D.2.4.1 — Market Risk Model. The market risk component is derived from VaR figures computed from the P&L distributions at a target confidence level of 99%.

Empirical Quantile. Where the target quantile falls between two observations, linear interpolation is applied. Let the P&L vector be sorted in ascending order with n observations; the quantile of the i -th sorted observation is:

Empirical Quantile

$$q(i) = 100\% \times \frac{i - 0.5}{n}$$

Interpolated VaR at confidence level α

$$\text{VaR}(\alpha) = - \left[\frac{\text{PL}(q^+) (q^+ - (1 - \alpha) \cdot 100) + \text{PL}(q^-) ((1 - \alpha) \cdot 100 - q^-)}{q^+ - q^-} \right]$$

q^+ and q^- are the nearest empirical quantiles above and below $(1 - \alpha) \times 100$ respectively. VaR is bounded from below at zero.

Robust VaR. To improve statistical robustness, VaR is first estimated at a lower confidence level (95% for filtered historical; 90% for stressed scenarios) and then scaled to the 99% target using quantiles of a heavy-tailed Student- t distribution:

Robust VaR Scaling Factor

$$\text{SF} = \frac{t_{\nu}^{-1}(\alpha_{\text{high}})}{t_{\nu}^{-1}(\alpha_{\text{low}})}$$

Robust VaR

$$\text{RVaR}(l, a) = \text{VaR}(\alpha_{\text{low}}, l, a) \times \text{SF}$$

Symbol	Description
α	Target confidence level (0.99)
α_{high}	Target confidence level (0.99)
α_{low}	Base confidence level (0.95 for filtered historical; 0.90 for stressed)
$t_{\nu}^{-1}(\alpha)$	Quantile (inverse CDF) of the Student- t distribution at confidence α with ν degrees of freedom
ν	Degrees of freedom of the Student- t distribution
$\text{RVaR}(l, a)$	Robust VaR for split l , scenario sub-sample a
n	Number of observations in the scenario sub-sample
q^+, q^-	Nearest empirical quantiles bracketing the target percentile

Market Risk Component Aggregation. Robust VaR is computed separately for each scenario sub-sample. The mean across sub-samples is taken within each scenario type. The final market risk component is the maximum of the filtered historical mean (plus the Correlation Break adjustment) and the stressed mean scaled by the stress weighting factor $w_s \in [0.60, 0.70]$:

Market Risk Component

$$\text{MRC}(l) = \max \left(\underbrace{\overline{\text{RVaR}}_{\text{hist}}(l) + \overline{\text{CB}}(l)}_{\text{filtered historical}}, \underbrace{w_s \cdot \overline{\text{RVaR}}_{\text{stress}}(l)}_{\text{stressed}} \right)$$

where $\overline{(\cdot)}$ denotes the mean across scenario sub-samples a .

Symbol	Description
$\text{MRC}(l)$	Market Risk Component for split l
$\overline{\text{RVaR}}_{\text{hist}}(l)$	Mean Robust VaR — filtered historical scenarios
$\overline{\text{RVaR}}_{\text{stress}}(l)$	Mean Robust VaR — stressed scenarios
$\overline{\text{CB}}(l)$	Mean Correlation Break Adjustment
w_s	Stressed scenario weighting factor (typically 0.60–0.70)

D.2.4.2 — Model Risk Buffer (Correlation Break Adjustment). A Correlation Break adjustment is applied to the market risk component to account for the risk that correlations shift adversely during a liquidation. The VaR is recalculated over rolling time-windows within the historical data; excess risk relative to the full-sample estimate is aggregated via a root-mean-square measure and clipped between a floor Φ and cap ψ (both expressed as fractions of $\text{RVaR}_{\text{hist}}$):

Excess Risk per Rolling Sub-Window T_j

$$\text{RVaR}^+(l, a, T_j) = \max(0, \text{RVaR}_{\text{sub}}(l, a, T_j) - \text{RVaR}_{\text{hist}}(l, a))$$

Root Mean Square Excess Risk

$$\text{RMSE}(l, a) = \sqrt{\frac{1}{n_a^+} \sum_{j=1}^{n_a^+} [\text{RVaR}^+(l, a, T_j)]^2}$$

Correlation Break Adjustment

$$\text{CB}(l, a) = \text{clip}\left(\kappa \cdot \text{RMSE}(l, a), \Phi \cdot \text{RVaR}_{\text{hist}}(l, a), \psi \cdot \text{RVaR}_{\text{hist}}(l, a)\right)$$

Symbol	Description
$\text{RVaR}_{\text{sub}}(l, a, T_j)$	Robust VaR over rolling window ending at T_j , sub-sample a
$\text{RVaR}_{\text{hist}}(l, a)$	Full sub-sample Robust VaR for split l , sub-sample a
$\text{RVaR}^+(l, a, T_j)$	Excess risk for rolling window ending at T_j
$\text{RMSE}(l, a)$	Root mean square of positive excess risk figures
κ	Calibration scaling factor
n_a^+	Count of sub-windows with positive excess risk
Φ	Floor as fraction of $\text{RVaR}_{\text{hist}}$ (minimum charge)
ψ	Cap as fraction of $\text{RVaR}_{\text{hist}}$ (maximum charge)
$\text{CB}(l, a)$	Correlation Break Adjustment for split l , sub-sample a

1.2.5 D.2.5 — Liquidity Risk Component

The liquidity risk component reflects the market impact and execution costs of liquidating positions in real market conditions. It is computed at the position level and then aggregated at the liquidation group split level with a diversification benefit applied.

Position - Level Liquidity Risk Component

$$LC(i) = \text{RVaR}_{\text{pos}}(i) \times \lambda(i)$$

The component scales with the positional market risk, reflecting that volatile instruments carry greater execution risk. $\lambda(i)$ is the instrument-level liquidity factor, capturing market impact relative to traded volume.

Diversification Factor

$$DF(l) = \max \left(DF_{\text{floor}}, \frac{\text{RVaR}_{\text{portfolio}}(l)}{\sum_{i \in l} \text{RVaR}_{\text{pos}}(i)} \right)$$

A well-diversified portfolio, where individual risks partially offset each other, receives a proportional reduction. The floor ensures a baseline liquidity charge is always present.

Aggregate Liquidity Risk Component

$$LC(l) = DF(l) \times \sum_{i \in l} LC(i)$$

Symbol	Description
$LC(i)$	Position-level Liquidity Risk Component
$\text{RVaR}_{\text{pos}}(i)$	Positional Robust VaR for instrument $i = \text{RVaR}_{\text{unit}}(i) \times N_i $
$\lambda(i)$	Liquidity factor for instrument i
$DF(l)$	Diversification Factor for split l
DF_{floor}	Minimum diversification factor
$\text{RVaR}_{\text{portfolio}}(l)$	Portfolio-level Robust VaR for split l
$LC(l)$	Aggregate Liquidity Risk Component for split l

1.2.6 D.2.6 — Aggregation and Final Initial Margin

Initial Margin — Liquidation Group Split

$$\text{IM}(l) = \text{MRC}(l) + LC(l)$$

Total Initial Margin — Customer Portfolio

$$\text{IM}_{\text{total}} = \sum_l \text{IM}(l)$$

No offsets are recognised across liquidation group splits.

Symbol	Description
$\text{IM}(l)$	Initial Margin for liquidation group split l
$\text{MRC}(l)$	Market Risk Component for split l
$LC(l)$	Liquidity Risk Component for split l
IM_{total}	Total portfolio Initial Margin

2. LTV Calculation Methodology (Section E)

2.1 E.2 — Collateral LTV: Haircuts on Posted Collateral

2.1.1 E.2.2 — Haircut Calibration

For each eligible collateral instrument, the haircut $H(i)$ is calibrated to cover the expected price decline over the collateral liquidation horizon at a 99% confidence level, using a historical simulation consistent with the initial margin scenario methodology. The resulting accepted value is the fraction of market value recognised as satisfying the margin obligation.

Collateral LTV

$$\text{LTV}_{\text{coll}}(i) = 1 - H_{\text{eff}}(i)$$

Accepted Collateral Value

$$\text{ACV}(i) = \text{MV}(i) \times (1 - H_{\text{eff}}(i))$$

Total Accepted Collateral Value

$$\text{ACV}_{\text{total}} = \sum_i \text{ACV}(i) + \sum_j \text{Cash}(j)$$

Cash posted in EUR is accepted at full value ($H = 0$). Non-EUR collateral carries an additional FX add-on applied to $H(i)$.

Symbol	Description
$H_{\text{eff}}(i)$	Haircut for collateral instrument i
$\text{MV}(i)$	Current market value of collateral instrument i
$\text{LTV}_{\text{coll}}(i)$	Collateral LTV (accepted fraction of market value)
$\text{ACV}(i)$	Accepted Collateral Value of instrument i
$\text{ACV}_{\text{total}}$	Total Accepted Collateral Value
$\text{Cash}(j)$	Cash posted in currency j

2.1.2 E.2.3 — Concentration Add-ons

Where a customer's collateral portfolio is concentrated within a correlated group of instruments, an additional concentration haircut is applied. The add-on is determined from the average pairwise correlation among instruments in each collateral asset group g :

Average Pairwise Correlation

$$\bar{\rho}(g) = \frac{2}{m(m-1)} \sum_{i < j, i, j \in g} \rho(i, j)$$

Computed over all pairs (i, j) within collateral asset group g ; m is the number of instruments in the group.

Diversification Score

$$\text{DS}(g) = \sqrt{\frac{1 + (m-1)\bar{\rho}(g)}{m}}$$

$\text{DS}(g) \rightarrow 1/\sqrt{m}$ for uncorrelated instruments; $\text{DS}(g) \rightarrow 1$ as $\bar{\rho}(g) \rightarrow 1$ (fully correlated). A fully uncorrelated portfolio receives no concentration add-on.

Concentration Add-on Haircut

$$H_{\text{conc}}(i, g) = H_{\text{base}}(i) \times \frac{\text{DS}(g) - \text{DS}_{\text{floor}}}{1 - \text{DS}_{\text{floor}}}$$

Applied as an increment to the base haircut where $\text{DS}(g) > \text{DS}_{\text{floor}}$. As average pairwise correlations rise, the diversification score falls and the add-on increases proportionally.

Effective Haircut

$$H_{\text{eff}}(i) = H_{\text{base}}(i) + H_{\text{conc}}(i, g)$$

Symbol	Description
$\rho(i, j)$	Pairwise return correlation between instruments i and j
$\bar{\rho}(g)$	Average pairwise correlation within collateral asset group g
m	Number of instruments in asset group g
$DS(g)$	Diversification Score for asset group g
DS_{floor}	Minimum diversification score (floor below which no add-on)
$H_{\text{base}}(i)$	Base haircut for collateral instrument i
$H_{\text{conc}}(i, g)$	Concentration add-on haircut for instrument i in group g
$H_{\text{eff}}(i)$	Effective total haircut for instrument i

2.2 E.3 — Loan-to-Value for Margin Lending and Leveraged Positions

2.2.1 E.3.2 — LTV Calculation

The LTV ratio measures the relationship between the outstanding loan exposure and the haircut-adjusted collateral value at any point in time. LTV ratios are calculated in real time using current market prices, with foreign currency positions converted to EUR at current spot rates.

Net Liquidation Value

$$NLV_t = MV_{\text{portfolio},t} - L_t$$

LTV Ratio

$$LTV_t = \frac{L_t}{ACV_{\text{total},t}}$$

An LTV ratio below 1.0 indicates collateral value exceeds the exposure. An LTV ratio at or above 1.0 triggers action.

Symbol	Description
NLV_t	Net Liquidation Value at time t
$MV_{\text{portfolio},t}$	Market value of all open positions at time t
L_t	Outstanding loan balance at time t
LTV_t	Loan-to-Value ratio at time t
$ACV_{\text{total},t}$	Total Accepted Collateral Value at time t ($= NLV_t + \text{additional posted collateral}$)

2.2.2 E.3.3 — Stress Testing LTV

In addition to the current LTV ratio, a stress LTV is calculated daily by applying the stressed scenario prices from the initial margin framework to both the financed positions and the collateral. It identifies positions at material risk of breaching margin breach limits under adverse market conditions.

Stress LTV

$$LTV_{\text{stress}} = \frac{L_t}{ACV_{\text{stress}}}$$

Stressed Accepted Collateral Value

$$ACV_{\text{stress}} = \sum_i MV_{\text{stress}}(i) \times (1 - H_{\text{eff}}(i)) + \sum_j \text{Cash}(j)$$

$MV_{\text{stress}}(i)$ uses scenario prices from the stressed scenario set of the initial margin framework.

Symbol	Description
LTV_{stress}	Stress Loan-to-Value ratio
ACV_{stress}	Accepted Collateral Value under stress scenario prices
$MV_{\text{stress}}(i)$	Market value of collateral instrument i at stressed scenario prices

A. Consolidated Formula Reference

This appendix lists all quantitative formulas from the methodology in the order in which they appear, with their canonical names.

Ref.	Name	Formula
D.1 — Variation Margin		
1	Variation Margin	$VM_{i,t} = (P_{i,t} - P_{i,t-1}) \times N_i$
D.2.2 — Scenario Construction		
2	Filtered (Volatility-Scaled) Return	$r_{\text{filt}}(i, s) = r_{\text{hist}}(i, s) \times \frac{\hat{\sigma}_{i,0}}{\hat{\sigma}_{i,s}}$
3	Filtered Historical Scenario Price	$P_{\text{filt}}(i, s) = P_{i,0} \times (1 + r_{\text{filt}}(i, s))$
4	Stressed Scenario Price	$P_{\text{stress}}(i, s) = P_{i,0} \times (1 + r_{\text{stress}}(i, s))$
D.2.3 — Profit and Loss Distributions		
5	Instrument P&L	$PL(i, s) = (P_{i,0} - P_{\text{scenario}}(i, s)) \times N_i$
6	Liquidation Group Split P&L	$PL(l, s) = \sum_{i \in l} PL(i, s)$
D.2.4.1 — Market Risk Model		
7	Empirical Quantile	$q(i) = 100\% \times \frac{i - 0.5}{n}$
8	Interpolated VaR	$VaR(\alpha) = - \left[\frac{PL(q^+)(q^+ - (1 - \alpha) \cdot 100) + PL(q^-)((1 - \alpha) \cdot 100 - q^-)}{q^+ - q^-} \right]$
9	Robust VaR Scaling Factor	$SF = \frac{t_{\nu}^{-1}(\alpha_{\text{high}})}{t_{\nu}^{-1}(\alpha_{\text{low}})}$
10	Robust VaR	$RVaR(l, a) = VaR(\alpha_{\text{low}}, l, a) \times SF$
11	Market Risk Component	$MRC(l) = \max(\overline{RVaR}_{\text{hist}}(l) + \overline{CB}(l), w_s \cdot \overline{RVaR}_{\text{stress}}(l))$
D.2.4.2 — Model Risk Buffer (Correlation Break Adjustment)		
12	Excess Risk per Sub-Window	$RVaR^+(l, a, T_j) = \max(0, RVaR_{\text{sub}}(l, a, T_j) - RVaR_{\text{hist}}(l, a))$
13	Root Mean Square Excess Risk	$RMSE(l, a) = \sqrt{\frac{1}{n_a^+} \sum_{j=1}^{n_a^+} [RVaR^+(l, a, T_j)]^2}$
14	Correlation Break Adjustment	$CB(l, a) = \text{clip}(\kappa \cdot RMSE(l, a), \Phi \cdot RVaR_{\text{hist}}(l, a), \psi \cdot RVaR_{\text{hist}}(l, a))$
D.2.5 — Liquidity Risk Component		
15	Position-Level Liquidity Charge	$LC(i) = RVaR_{\text{pos}}(i) \times \lambda(i)$
16	Diversification Factor	$DF(l) = \max\left(DF_{\text{floor}}, \frac{RVaR_{\text{portfolio}}(l)}{\sum_{i \in l} RVaR_{\text{pos}}(i)}\right)$
17	Aggregate Liquidity Risk Component	$LC(l) = DF(l) \times \sum_{i \in l} LC(i)$
D.2.6 — Aggregation and Final Initial Margin		
18	Initial Margin (Split)	$IM(l) = MRC(l) + LC(l)$

Continued on next page

Ref.	Name	Formula
19	Total Initial Margin	$IM_{\text{total}} = \sum_l IM(l)$
E.2.2 — Haircut Calibration		
20	Collateral LTV	$LTV_{\text{coll}}(i) = 1 - H(i)$
21	Accepted Collateral Value	$ACV(i) = MV(i) \times (1 - H(i))$
22	Total Accepted Collateral Value	$ACV_{\text{total}} = \sum_i ACV(i) + \sum_j \text{Cash}(j)$
E.2.3 — Concentration Add-ons		
23	Average Pairwise Correlation	$\bar{\rho}(g) = \frac{2}{m(m-1)} \sum_{i < j, i, j \in g} \rho(i, j)$
24	Diversification Score	$DS(g) = \sqrt{\frac{1 + (m-1) \bar{\rho}(g)}{m}}$
25	Concentration Add-on Haircut	$H_{\text{conc}}(i, g) = H_{\text{base}}(i) \times \frac{DS(g) - DS_{\text{floor}}}{1 - DS_{\text{floor}}}$
26	Effective Haircut	$H_{\text{eff}}(i) = H_{\text{base}}(i) + H_{\text{conc}}(i, g)$
E.3 — Loan-to-Value for Margin Lending		
27	Net Liquidation Value	$NLV_t = MV_{\text{portfolio}, t} - L_t$
28	LTV Ratio	$LTV_t = \frac{L_t}{ACV_{\text{total}, t}}$
29	Stress LTV	$LTV_{\text{stress}} = \frac{L_t}{ACV_{\text{stress}}}$
30	Stressed Accepted Collateral Value	$ACV_{\text{stress}} = \sum_i MV_{\text{stress}}(i) \times (1 - H_{\text{eff}}(i)) + \sum_j \text{Cash}(j)$

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B. Worked Example — Customer Opens a 3× Leveraged Position

The following example illustrates the end-to-end margin calculation for a customer wishing to open a 3× leveraged position in an equity ETF.

Mock Data

Parameter	Value
Instrument	MSCI World ETF (IE00B4L5Y983)
Current price P_0	€100.00 per share
Quantity N	150 shares
Position value	$MV_0 = 150 \times €100 = €15,000$
Target leverage	3×
Customer equity	$€15,000/3 = €5,000$
Loan L_0	$€15,000 - €5,000 = €10,000$
Annualised vol σ	18%
Asset-class haircut floor h_{floor}	10% (equity ETF)
Volatility buffer weight w_{buf}	0.50
Haircut horizon	1 trading day

Step 1 — Collateral Assessment (Phase 1 / Section E.2)

Volatility-implied haircut over a 1-day horizon at 99%:

$$h_{\text{vol}} = 1 - e^{-z_{0.99} \sigma \sqrt{1/252}} = 1 - e^{-2.326 \times 0.18 \times 0.0630} \approx 2.6\%$$

Since $h_{\text{vol}} = 2.6\% < h_{\text{floor}} = 10\%$, the floor binds and the vol excess is zero:

$$h_{\text{base}} = 10\% + 0.5 \times \max(0, 2.6\% - 10\%) = 10\%$$

No concentration add-on ($h_{\text{conc}} = 0$), so $h_{\text{eff}} = 10\%$.

$$\text{ACV}_0 = \text{€}15,000 \times (1 - 0.10) = \text{€}13,500$$

Step 2 — Initial Margin Calculation (Phase 2 / Section D.2)

From 250 filtered historical scenarios (volatility-scaled, no Student-t anchor):

$$\text{RVaR}^{\text{hist}} \approx 5.9\% \quad (99\text{th percentile loss over 5-day liquidation horizon})$$

$$\text{CB} = 0.5\% \Rightarrow \text{MRC} = 5.9\% + 0.5\% = 6.4\%$$

$$\text{LC} = \text{DF} \times \text{RVaR}^{\text{hist}} \times \lambda = 0.85 \times 5.9\% \times 1.0 = 5.0\%$$

$$\text{IM}_{\text{rate}} = 6.4\% + 5.0\% = 11.4\% \Rightarrow \text{IM}_{\text{EUR}} = 11.4\% \times \text{€}15,000 = \text{€}1,710$$

Step 3 — Trade Execution Gate (Phase 3)

Check	Value	Threshold	Result
Customer equity vs. IM	€5,000 ≥ €1,710	IM _{EUR}	Pass
ACV vs. IM	€13,500 ≥ €1,710	IM _{EUR}	Pass
Opening LTV	€10,000/€13,500 = 74.1%	75%	Pass

All checks pass. **Position is opened.**

$$\text{LTV}_0 = \frac{L_0}{\text{ACV}_0} = \frac{\text{€}10,000}{\text{€}13,500} = 74.1\%$$

$$\text{Max leverage at LTV} = 75\% : L^* = \frac{1}{0.25 + 0.75 \times 0.10} = \frac{1}{0.325} = 3.08 \times \quad \checkmark$$

Step 4a — Day 1: Adverse Price Move (Phase 4 / Section D.1.3)

Price falls from €100 to €97 (−3%):

$$\text{VM}_1 = (97 - 100) \times 150 = -\text{€}450 \quad (\text{debited from customer account})$$

$$\text{MV}_1 = 97 \times 150 = \text{€}14,550 \quad \text{ACV}_1 = \text{€}14,550 \times 0.90 = \text{€}13,095$$

$$\text{LTV}_1 = \frac{\text{€}10,000}{\text{€}13,095} = 76.4\% > 75\% \Rightarrow \text{Monitoring alert triggered}$$

Step 5a — Day 1: LTV Limit Check (Phase 5 / Section G.3.4)

Threshold	Limit	LTV ₁	Action
Monitoring	75%	76.4%	Flag — enhanced surveillance
Margin breach	80%	76.4%	No action
Liquidation	90%	76.4%	No action

Step 4b — Day 2: Price Recovery (Phase 4)

Price recovers from €97 to €99 (+2.1%):

$$\text{VM}_2 = (99 - 97) \times 150 = +\text{€}300 \quad (\text{credited to customer account})$$

$$\text{MV}_2 = 99 \times 150 = \text{€}14,850 \quad \text{ACV}_2 = \text{€}14,850 \times 0.90 = \text{€}13,365$$

$$\text{LTV}_2 = \frac{\text{€}10,000}{\text{€}13,365} = 74.8\% < 75\% \Rightarrow \text{Monitoring flag cleared — position healthy}$$

Summary

Day	Price	VM	ACV	LTV	
$t = 0$ (open)	€100.00	—	€13,500	74.1%	OK
$t = 1$	€97.00	–€450	€13,095	76.4%	Monitoring
$t = 2$	€99.00	+€300	€13,365	74.8%	OK

The example demonstrates that a $3\times$ leveraged position in this instrument opens comfortably within the 75% LTV limit and that daily Variation Margin adjustments provide a dynamic early-warning mechanism. A single adverse 3% daily move is sufficient to trigger the monitoring threshold, highlighting the sensitivity of leveraged positions to price changes and the importance of the ongoing maintenance margin framework.

Extended Example — Monitoring Framework Across Liquidation Horizon

The following extends the worked example above to illustrate the two-phase monitoring framework (Section D.1.5), including the loan evolution through Variation Margin, the overnight haircut check, and the transition to the seasoned phase at $t = 5$.

	Parameter	Value
Setup.	Opening price P_0	€100.00
	Quantity N	150 shares
	Position value MV_0	€15,000
	Opening loan L_0	€10,000
	Intraday haircut h_{eff}	10%
	Overnight haircut $h_{\text{overnight}}$	14%
	Monitoring threshold $\overline{LTV}$	75%
	Liquidation threshold LTV_{liq}	90%
	Liquidation horizon T_{liq}	4 trading days

Step 1 — Collateral Assessment.

$$h_{\text{vol}} = 1 - e^{-2.326 \times 0.18 \times \sqrt{1/252}} \approx 2.6\% < h_{\text{floor}} = 10\% \Rightarrow h_{\text{eff}} = 10\%$$

$$ACV_0 = €15,000 \times (1 - 0.10) = €13,500$$

Step 2 — Initial Margin.

$$RVaR^{\text{hist}} \approx 5.9\%, \quad CB = 0.5\%, \quad MRC = 6.4\%, \quad LC = 5.0\%$$

$$IM_{\text{rate}} = 11.4\% \Rightarrow IM_{\text{EUR}} = €1,710$$

Step 3 — Execution Gate.

$$LTV_0 = \frac{€10,000}{€13,500} = 74.1\% < 75\% \quad \checkmark \quad \text{Position opened.}$$

Step 4 — Day 1: Price –3% to €97 (Phase 1 — within liquidation horizon).

$$VM_1 = (97 - 100) \times 150 = -€450 \quad L_1 = €10,000 - (-€450) = €10,450$$

$$ACV_1 = €14,550 \times 0.90 = €13,095 \quad LTV_1 = \frac{€10,450}{€13,095} = 79.8\% > 75\%$$

Monitoring alert triggered (IM + VM breach, Phase 1).**Step 5 — Day 1 End-of-Day: Overnight Check.**

$$ACV_1^{\text{overnight}} = €14,550 \times (1 - 0.14) = €12,513 \quad LTV_1^{\text{overnight}} = \frac{€10,450}{€12,513} = 83.5\% > 75\%$$

Overnight margin review triggered. Customer must post additional collateral or reduce the loan before next open. $83.5\% < 90\%$: no liquidation.

Step 6 — Day 2: Price +2% to €99.

$$VM_2 = (99 - 97) \times 150 = +€300 \quad L_2 = €10,450 - €300 = €10,150$$

$$ACV_2 = €14,850 \times 0.90 = €13,365 \quad LTV_2 = \frac{€10,150}{€13,365} = 75.9\% > 75\%$$

Monitoring flag remains active; no overnight breach ($LTV_2^{\text{overnight}} < 75\%$ if overnight haircut does not push ACV below threshold).

Step 7 — Day 5: Position Seasoned ($t > T_{liq}$), Price stable at €99. Phase 2 now applies. Initial Margin is no longer binding; Maintenance Margin governs.

$$LTV_5 = \frac{€10,150}{€13,365} = 75.9\% > 75\% \Rightarrow \text{MM monitoring alert (Phase 2).}$$

Customer must top up collateral or reduce the loan to clear the flag.

	Day	Price	VM	Loan	ACV	LTV / Status
Summary.	$t = 0$ (open)	€100	—	€10,000	€13,500	74.1% OK
	$t = 1$	€97	−€450	€10,450	€13,095	79.8% <i>Monitoring (IM+VM)</i>
	$t = 1$ EOD	€97	—	€10,450	€12,513 [†]	83.5% <i>Overnight review</i>
	$t = 2$	€99	+€300	€10,150	€13,365	75.9% <i>Monitoring</i>
	$t = 5+$	€99	—	€10,150	€13,365	75.9% <i>MM monitoring (Phase 2)</i>

[†]Overnight ACV uses $h_{\text{overnight}} = 14\%$.

Appendix: Comparison with Interactive Brokers Margin Framework

This appendix applies the Interactive Brokers (IB) margin framework to the identical worked example used in the preceding sections — 150 shares at €100 per share, yielding a €15,000 position with €5,000 customer equity and a €10,000 loan (3× leverage) — to facilitate a direct methodological comparison.

IB Approach A: Regulation T (Reg T)

Reg T is a flat-rate rule issued by the US Federal Reserve (12 CFR Part 220) and enforced by FINRA Rule 4210. It defines a minimum initial margin of 50% of the purchase price for long equity positions, irrespective of the actual risk profile of the instrument.

Step 1 — Compute the Reg T initial margin requirement.

$$IM_{\text{RegT}} = 50\% \times €15,000 = €7,500$$

Step 2 — Assess whether the position can be opened. Customer equity is €5,000, which is less than the required €7,500. **The €15,000 position cannot be opened under Reg T.** The same €5,000 of equity permits a maximum position of:

$$MV_{\text{max}} = \frac{€5,000}{50\%} = €10,000 \implies 100 \text{ shares at } €100 \quad (2\times \text{leverage only})$$

Step 3 — Adjusted position (maximum under Reg T).

Item	Value
Shares	100
Market value	€10,000
Loan	€5,000
Customer equity	€5,000
Leverage	2×
Reg T initial margin rate	50%

Step 4 — Maintenance margin requirement (ongoing). FINRA Rule 4210 sets a minimum maintenance margin of 25% of current market value. IB applies this floor without further add-ons for standard large-cap equities:

$$MMR = 25\% \times €10,000 = €2,500$$

$$\text{Excess Liquidity} = \text{Equity} - MMR = €5,000 - €2,500 = €2,500$$

IB begins forced liquidation when Excess Liquidity drops below zero, without issuing a margin call. There is no graduated warning framework.

Step 5 — Compute the auto-liquidation price. Let $m = 0.25$ (maintenance rate), $L = €5,000$, $N = 100$ shares:

$$P^* = \frac{L}{N \times (1 - m)} = \frac{€5,000}{100 \times 0.75} = €66.67$$

The stock must fall −33.3% from €100 before IB triggers auto-liquidation.

Step 6 — IB account panel (at position open).

IB Field	Formula	Value
Net Liquidation Value	Equity	€5,000
Initial Margin Req.	$50\% \times MV$	€5,000
Maintenance Margin Req.	$25\% \times MV$	€2,500
Available Funds	$\text{Equity} - \text{IMR}$	€0 (at limit)
Excess Liquidity	$\text{Equity} - \text{MMR}$	€2,500
Cushion	$\text{Excess Liq.} / \text{NLV}$	50%

IB Approach B: Portfolio Margin (PM)

Portfolio Margin is available to IB accounts with net liquidation value above \$110,000 (FINRA Rule 4210(g)). It replaces the flat Reg T rule with a scenario-based engine derived from the OCC's Theoretical Inter-market Margin System (TIMS).

Step 1 — Run the OCC scenario grid. IB applies 31 price-shock scenarios ranging from −15% to +15% in 1% steps, combined with implied-volatility shocks of ±75% at each price point. For a plain long equity position the dominant scenario is the −15% price shock:

$$\text{Scenario loss} = 15\% \times €15,000 = €2,250$$

Step 2 — Apply the single-stock PM floor. FINRA Rule 4210(g) mandates a minimum requirement of 15% of market value for individual equity positions:

$$PM_{\text{floor}} = 15\% \times \text{€}15,000 = \text{€}2,250$$

$$PM_{\text{req}} = \max(\text{scenario loss}, PM_{\text{floor}}) = \text{€}2,250$$

Step 3 — Assess whether the position can be opened. Customer equity €5,000 > €2,250 required. **The €15,000 position can be opened under PM (3× leverage is feasible).**

Step 4 — Maximum leverage under PM.

$$\text{Max leverage} = \frac{1}{15\%} \approx 6.7\times \implies MV_{\text{max}} = \frac{\text{€}5,000}{15\%} = \text{€}33,333$$

Step 5 — PM maintenance and liquidation trigger. The scenario grid is re-evaluated continuously throughout the trading day. For single equities, the maintenance requirement equals the PM requirement (15% floor). Forced liquidation is triggered when Excess Liquidity (NLV – PM_req) drops below zero.

Side-by-Side Comparison

The table below compares all three frameworks on the €15,000 worked example. Where IB Reg T requires an adjusted 2× position, figures are shown on a comparable per-EUR basis.

Dimension	IB Reg T	IB Portfolio Margin	Trade Republic
Methodology	Flat regulatory rule (50%)	OCC scenario grid ±15%, 31 points	Filtered historical simulation + EWMA volatility scaling
Initial margin (€15,000 position) 3× leverage possible?	€7,500 (50%) No — needs €7,500 equity	€2,250 (15%) Yes	€1,710 (11.4%) Yes
Max leverage on €5,000 equity	2×	6.7×	~8× (LTV-capped at 75%)
IM rate driver	Regulatory flat rate	Worst-case ±15% scenario + 15% floor	99% RVaR over 5-trading-day horizon
Maintenance trigger	Excess Liquidity < 0 (binary)	Excess Liquidity < 0 (binary)	LTV > 90% (liquidation); LTV > 75% (monitoring)
Graduated warning framework	None	None	Yes — three thresholds (75% / 80% / 90%)
Auto-liquidation price (3×, €100)	N/A (not permitted)	$P^* \approx \text{€}83$ at 15% MM	$LTV = 90\% \implies P^* \approx \text{€}74$
Concentration add-on	ADV-based escalation (not published)	OCC concentration rules	$h_{\text{conc}} + 5\%$ when weight > 20%
Liquidation horizon	No explicit horizon	1-day (intraday scenarios)	5 trading days
Intraday vs. overnight distinction	Yes — overnight reverts to 50% Reg T	Yes — overnight PM re-evaluated at 15:50 ET	Yes — overnight haircut add-on applied at EOD
Margin call issued	No — auto-liquidates	No — auto-liquidates	No — auto-liquidates

Key Observations

- Reg T is purely rule-based.** The 50% flat rate bears no relation to the actual volatility or liquidity of the instrument. A low-vol blue-chip and a speculative small-cap receive identical treatment. This blocks 3× leverage entirely, regardless of the risk profile.
- IB Portfolio Margin is methodologically closest to the Trade Republic framework.** Both apply scenario-based stress tests. IB uses the OCC's fixed ±15% grid; Trade Republic uses filtered historical simulation calibrated to the realised tail of each ISIN over the trailing 750 trading days.

3. **The Trade Republic IM is lower than the IB PM floor** (11.4% vs. 15%) because the historical approach captures the actual volatility of the stock rather than applying a fixed 15% shock. For a low-volatility instrument the TR rate will be materially lower; for a high-volatility instrument it will exceed the IB PM floor.
4. **The liquidation frameworks differ in gradualism.** IB operates a binary trigger (Excess Liquidity $< 0 \Rightarrow$ liquidate). Trade Republic operates a three-level graduated framework (monitoring at 75%, breach at 80%, liquidation at 90%), giving the customer explicit visibility and response time before forced liquidation occurs.
5. **Both IB and Trade Republic auto-liquidate** without issuing a traditional margin call. The difference is that Trade Republic's monitoring thresholds function as early warnings that a formal margin call framework would otherwise provide.